ME210L: Theory of Mechanisms and Machines (Spring 2025)

Project submission: Report

Group number: 15

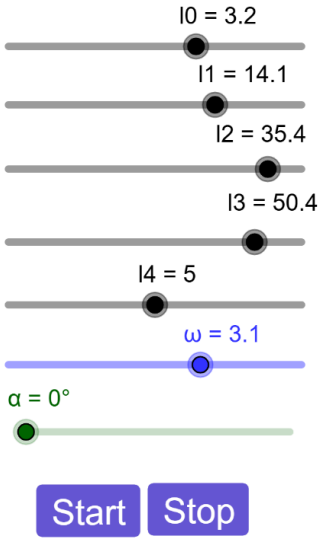
Title: Windscreen Wiper Mechanism

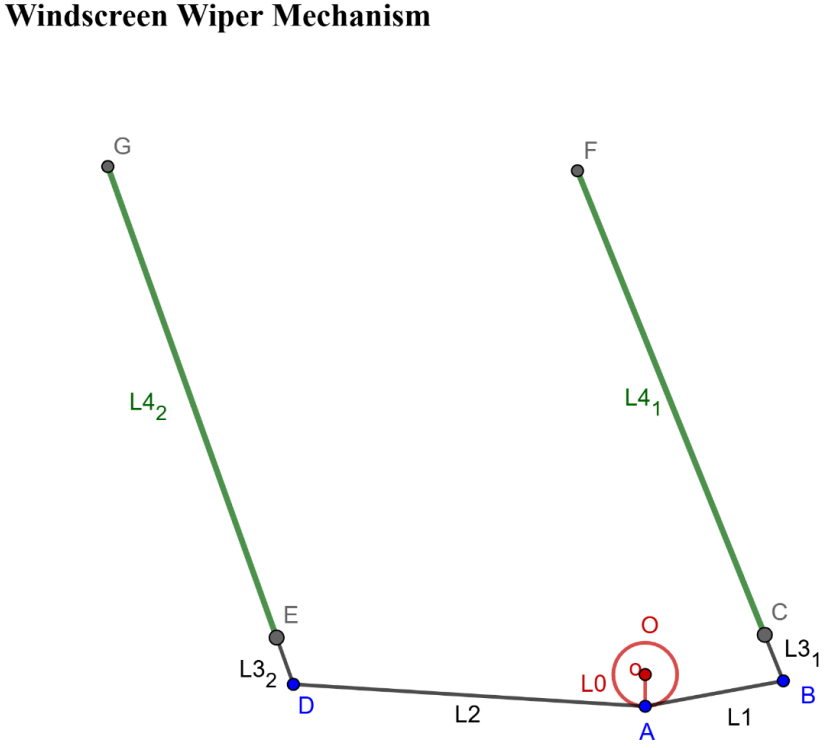
Aim: To obtain the working of the windscreen wiper mechanism and carry out its velocity, acceleration, locate instantaneous centers, and force analysis.

Software used: GeoGebra

Description (include a sketch): The windscreen wiper mechanism invention dates back to 1896, being its only intended use for rapidly moving vehicles. As years pass, this mechanism evolves and several variants come about. The first manual windscreen mechanism was invented in 1903 where the driver needs to rotate the lever. In 1919, the first automatic windshield wiper was created.In 1926, Bosch introduced its invention - an electric windshield wiper. The wiper mechanism rotated using plain bearings and was driven by a small electric motor connected to the car's battery, independent of the engine's electrical system. The main reason behind the invention of this mechanism to clean off the dust/water from the windshield of an automobile. This link-age mechanism is an example of a crank-rocker mechanism.

Kinematic working of the mechanism:
There are three joints that are grounded, those being, 'O', 'C', 'E'. 'O' is where the motor is located, the motor applies a constant torque and the crank rotates in a counterclockwise direction. The crack 'OA', is rotated with a uniform angular velocity, 'ω'. Due to the crank, the couplers, 'BA' and 'DA' rotate in a counterclockwise motion as well with 'ω_1' and 'ω_2' angular velocities. As the couplers rotate, the links 'CB' and 'DE' oscillate as rockers.



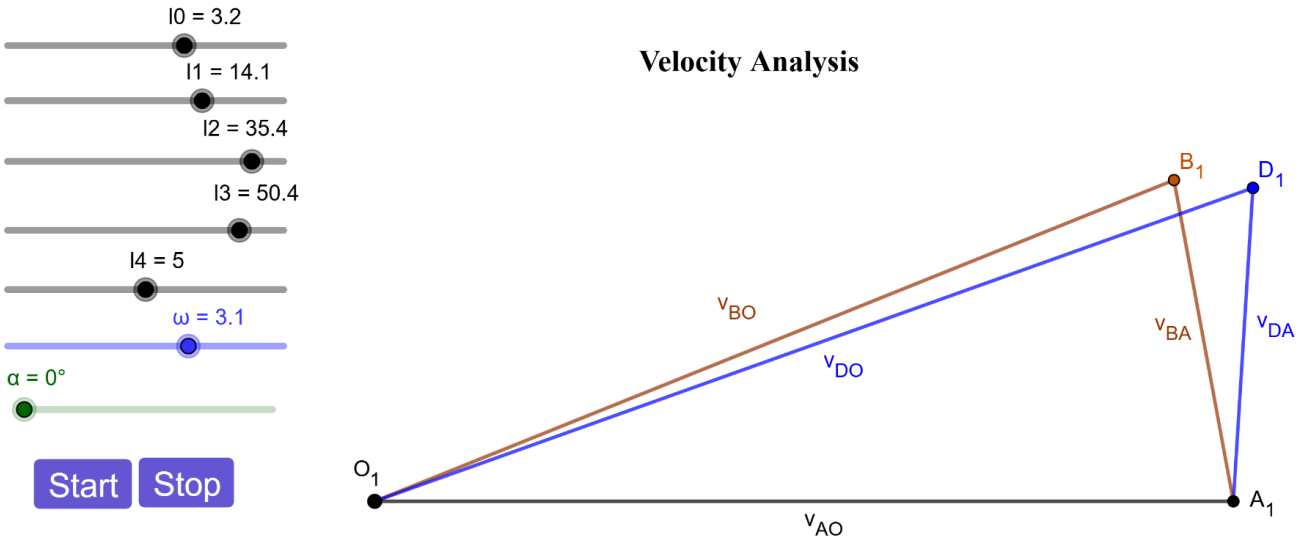


(note: 1. L3 corresponds to l4 and L4 corresponds to l3, the rest are simply their counterparts as L0=l0.)

Velocity Analysis: Velocity follows the formula; $v = (\omega * R)$

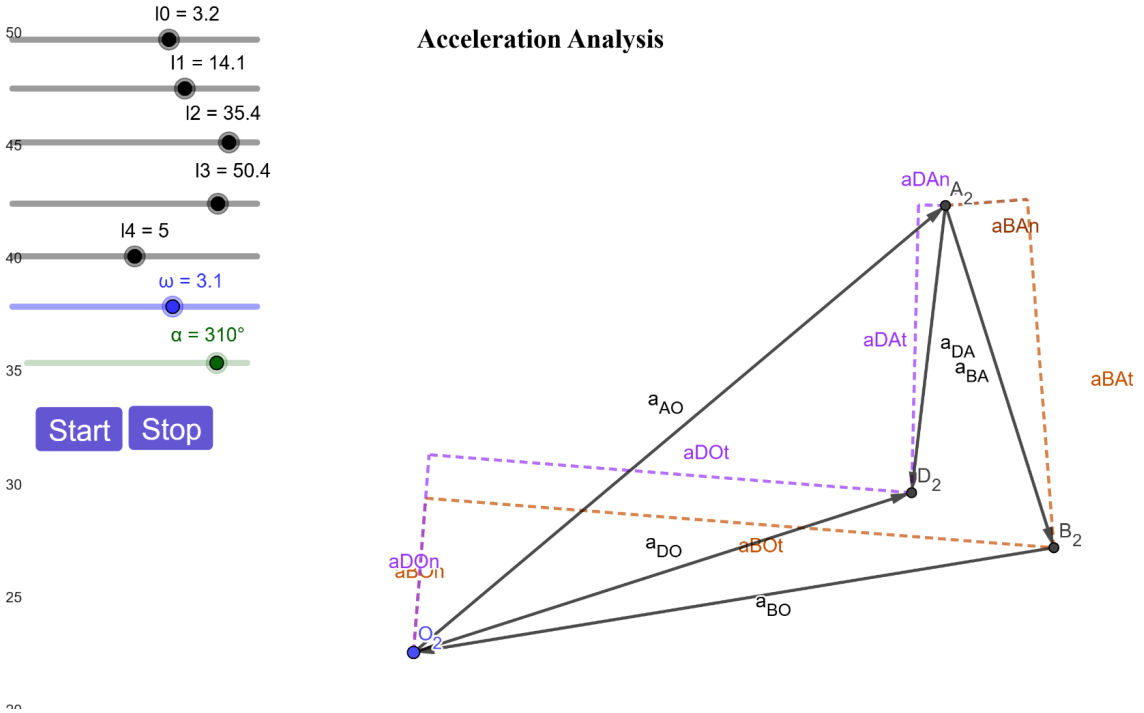
The crank and coupler 's velocity is determined by the angular velocity 'ω' and the length of link L0 as; $v_{AO} = (\omega * LO)$.

Using the Graphical method, first we draw v_{AO} perpendicular from AO. we can just draw the perpendiculars BA and BC. Intersection of the perpendiculars from BA and BC is B_1. Similarly we find the D_1 by finding the intersection of the perpendiculars of DA and DE. Join B1 and D1 with O1, and then with A1.



Acceleration Analysis: First find the angular velocities of links L1, L2, $L3_1$, and $L3_2$ using the formulas; $(\omega_1 = v_{BA}/l1)$, $(\omega_2 = v_{DA}/l2)$, $(\omega_3 = v_{BO}/l4)$, and $(\omega_4 = v_{DO}/l4)$.

Having the angular velocities, we can find acceleration in the normal direction, 'a_n'. Tangential acceleration 'a_t' is perpendicular to the a_n. The crank is moving in a uniform velocity, this means there is no acceleration in the tangential direction for 'AO'. Using the formula; $(a_n = \omega^2 * R)$, we can find a_{AOn}, a_{BOn}, a_{BAn}, a_{DOn}, and a_{DAn}. Drawing the perpendiculars of the normal accelerations, the intersection point is B_2 for OAB, and then for OAD it is D_2.



Links: There are 6 links in total (L=6). The types of links present are all 'First Order'.
Crank is L0 link, OA. There are two couplers which are the L1 and L2 links representing 'BA' and 'DA' respectively. 'BC' and 'DE' are rockers. Another link is ground.

Joints: In total there are 7 joints, they mainly consist of revolute joints. Motor Shaft to Crack is considered to be a single rigid joint. Likewise crack-to-coupler, coupler-to-fixed frame, and fixed frame-to-wiper arm all have single revolute joints. In this mechanism, the fixed frame-to-wiper arm joint is used as a rigid joint.
Mobility Calculation:

$$M = 3(L - 1) - 2J_1 - J_2$$

$$M = 3(6 - 1) - 2(7) = 1$$

Mobility/Degrees of Freedom of the Mechanism is 1.

Comparison of Windscreen Wiper Mechanisms

Mechanism Type	Working	Advantages	Disadvantages
Parallel Arm Mechanism	Both wipers move in the same direction simultaneously. Common in buses, trucks, and high-end cars. Provides good coverage but requires a complex linkage	- Covers a wide area efficiently - Works well for large windshields	- More complex linkage - Can be expensive to maintain
Opposed Arm Mechanism	Wipers move in opposite directions towards each other. Found in most modern cars.	- Provides symmetrical wiping coverage - Simple and effective design	- Can leave small unwiped areas near edges
Single Arm Mechanism	Uses a single wiper arm sweeping across the windshield.	- Minimalist design - Some have extending arms for better coverage	- May not cover the full windshield - Less stable in high speeds

Pantograph Mechanism	Uses a dual-arm system for stability and uniform wiping. Mostly used in trains, buses, and large vehicles.	- Ensures uniform pressure across the windshield - High stability	- Complex and expensive - Requires strong motors
Radial (Pendulum) Mechanism	Wiper pivots from a single fixed point. Found in older cars and some budget vehicles. May leave portions of the windshield unwiped.	- Simple and cost-effective - Fewer moving parts	- Leaves large unwiped areas, especially on larger windshields
Cam & Lever Mechanism	Uses a cam-driven system to control wiper motion. Less common, used in specialized vehicles.	- Precise movement control - Can be optimized for specific shapes	- Less commonly used - Requires precise engineering
Hinged Wiper Mechanism	This mechanism has a hinged joint allowing angle adjustments dynamically.	- Adjusts dynamically to windshield shape - Improves wiping efficiency	- More moving parts can lead to wear and tear

Capabilities of your mechanism model:

Inputs: Rotational motion from the motor - (i.e. angular velocity 'ω' and angle ‘α’)
Outputs: Oscillatory motion of the wiper blades, velocity and acceleration of links, and location of instantaneous centers.
Adjustable Parameters: Angle of movement, angular velocity of the crank, lengths of every link.

Practical uses of the mechanism:

1. **Automobile windscreen wipers (Automotive and Aviation Use):**

Rain, snow, dirt and debris removal: Removes these particles from the automobile’s screen to improve visibility.
Headlight cleaning: Some modern cars have windscreen wipers for headlights to improve visibility at night during inclement weather.
Aircraft Windshields: In aviation, this mechanism is also used to clean the cockpit's windshield.
Aircraft De-icing: scrapes of the ice/frost from wings, windows, and other surfaces to prevent buildup.

2. **Automated Cleaning Systems for Other Surfaces:**

Solar panels: A windscreen wiper mechanism can be adapted for solar panel cleaning ensuring maximum energy efficiency.
Signage and Billboards: For large outdoor signs or billboards, windscreen wiper mechanism is used so that they are visible for all weather conditions.
Other areas where we can use this mechanism are in; cleaning system of high-rise buildings, cleaning of surfaces during industrial inspection, self-cleaning glasses in homes and buildings, and cleaning debris from machinery during manufacturing. Cleaning systems are also deployed in semiconductors as well.

3. **Agricultural Use**

Greenhouse Ventilation: In large greenhouses, a windscreen wiper mechanism can be used to clear water or condensation from glass panes to maintain airflow and visibility inside the greenhouse.
Tractor Windshield Wipers: For farm vehicles when working in dusty or muddy environments.

4. **Automated Coating System**

This mechanism helps to smooth or level out the coating when it is applied to a surface.

5. **Removal of Excess Material/Particles**

In film development, the mechanism is used to remove the extra chemicals/developer.

The windscreen wiper mechanism is used in many areas but usually for the same purpose which is cleaning/removal of debris to improve visibility.

Code:

```
%% Windscreen-wiper: Dynamic analysis
clc
%%Inputs
param.L1 =input("enter L1 in mm");
param.L2 = input("enter L2");
param.L3 =input("enter L3");
param.L4 =input("enter L4");  %mm
%%param.lp2=0;
param.th1 =0;
param.th2 = input("enter th2 in rad"); % rad
param.w1 = 0;
param.w2 = input("enter w2 in rad/s");  % rad/s
param.alp2=input("enter alp2 in rad/s^2"); %rad/s^2
param.m1 = input("enter m1 in kg ");
param.m2 = input("enter m2");
param.m3 = input("enter m3");
param.m4 = input("enter m4"); % kg
param.T_in=input("enter T_in"); %in N.m
param.J1= param.m1*param.L1^2/12;
param.J2 = param.m2*param.L2^2/12;
param.J3 = param.m3*param.L3^2/12;
param.J4 = param.m4*param.L4^2/12;
%param.J5 = param.m5*param.L5^2/12;  % kgm2
param.g = 9.81; % m/s2

%% Position analysis
fun = @(X) pos_solve(X,param);
options = optimoptions('fsolve','display','iter-detailed','Algorithm','levenberg-marquardt');
X0=[50/180*pi;4];
[X_solved] = fsolve(fun,X0,options);
param.th3=2*pi-X_solved(1);
param.th4=2*pi-X_solved(2);


%% Velocity analysis
fun = @(X)vel_solve(X,param);
options = optimoptions('fsolve','Algorithm','levenberg-marquardt');
X0=[20/180*pi;3];
[X_solved] = fsolve(fun,X0,options);
param.w3=X_solved(1);
param.w4=X_solved(2);

%% Acceleration analysis
fun = @(X)acc_solve(X,param);
options = optimoptions('fsolve','Algorithm','levenberg-marquardt');
X0=[20/180*pi;3];
[X_solved] = fsolve(fun,X0,options);
param.alp3=X_solved(1);
param.alp4=X_solved(2);

%% Accelerations of centroids
w2vector=[0;0;param.w2];
```

```
w4vector=[0;0;param.w4];
R_A=param.L2*[cos(param.th2);sin(param.th2);0];
alp2vector=[0;0;param.alp2];
R_B=param.L4*[cos(param.th4);sin(param.th4);0];
alp4vector=[0;0;param.alp4];

% Acceleration of A
a_An=cross(w2vector,cross(w2vector,R_A));
a_At=cross(alp2vector,R_A)
a_A=a_An+a_At;
param.a_G2=a_A/2;

% Acceleration of B
a_Bn=cross(w4vector,cross(w4vector,R_B));
a_Bt=cross(alp4vector,R_B);
```

Position

```
function [res] = pos_solve(X,param)
th3=X(1);
th4=X(2);
R_O4=param.L1;
R_A=param.L2*[cos(param.th2);sin(param.th2);0];
R_B=param.L4*[cos(th4);sin(th4);0];
R_B_A=param.L3*[cos(th3);sin(th3);0];
residue=R_B_A+R_A-R_B-R_O4;
res=residue(1:2);
end
```

Velocity

```
function [res] = vel_solve(X,param)
w3=X(1);
w4=X(2);
%input
R_A=param.L2*[cos(param.th2);sin(param.th2);0];
w2vector=[0;0;param.w2];
V_A=cross(w2vector,R_A);
R_B=param.L4*[cos(param.th4);sin(param.th4);0];
w4vector=[0;0;w4];
V_B=cross(w4vector,R_B);
R_B_A=param.L3*[cos(param.th3);sin(param.th3);0];
w3vector=[0;0;w3];
V_B_A=cross(w3vector,R_B_A);
residue=V_B_A-V_B+V_A;
res=residue(1:2);
end
```

Acceleration

```
function [res] = acc_solve(X,param)
alp3=X(1);
alp4=X(2);
alp3vector=[0;0;alp3];
```

```
alp4vector=[0;0;alp4];
alp2vector=[0;0;param.alp2];
R_A=param.L2*[cos(param.th2);sin(param.th2);0];
w2vector=[0;0;param.w2];
R_B=param.L4*[cos(param.th4);sin(param.th4);0];
w4vector=[0;0;param.w4];
R_B_A=param.L3*[cos(param.th3);sin(param.th3);0];
w3vector=[0;0;param.w3];
a_B_An=cross(w3vector,cross(w3vector,R_B_A));
a_B_At=cross(alp3vector,R_B_A);
a_Bn=cross(w4vector,cross(w4vector,R_B));
a_Bt=cross(alp4vector,R_B);
a_An=cross(w2vector,cross(w2vector,R_A));
a_At=cross(alp2vector,R_A);
residue=(a_B_An+a_B_At)-(a_Bn+a_Bt)+(a_An+a_At);
res=residue(1:2);
end
```

Results and Inferences:

- The model shows that this crank-rocker mechanism converts a continuous rotary motion into a limited oscillatory motion.
- The model gives us the velocities and the accelerations of the mechanism.
- The speed of the linkage system depends on the length of the crank, coupler, and rockers.
- This design ensures that there is adequate wiping coverage with minimal torque needed. If angular velocity changes by a bit, then there is massive change in the speed of the mechanism.
- Only the crank moves due to applied torque from the motor, the rest of the links are dependent on the crank. This means once the mechanism is built into an automobile or any other intended uses, we can't change the lengths but we can change the torque applied (i.e. angular velocity).
- This system is fairly simple due to the above fact, this rendering it easy to change and make variations of it. Hence there are many different types of windscreen wipers.

Online Link to Model: https://www.geogebra.org/classic/wse73spk